

ORIGINAL ARTICLE

Care Models for the Genetic Evaluation of Dilated Cardiomyopathy at Sites of the DCM Consortium

Elizabeth Jordan¹ ID, MMSc; Tia Moscarello¹ ID, MS; Hibatallah Khafagy, BS; Patricia K. Parker¹ ID, MS; Phoenix Grover, MS; Simone Weinmann, MS; Joseph Liu¹ ID, MS; Alberta Nomo, BA; Naomi Barker¹ ID, MS; Emily E. Brown¹ ID, MGC; Akos Berthold¹ ID, MMSc; Jessica Chowns, MS; Susan Christian¹ ID, PhD; Amy Ekwurtzel, MMS; Judy Fan, MS; Monisha Kisling, MS; Daria Ma, MS; Erin M. Miller¹ ID, MS; Jessica Sweeney, ScM; Brian Reys¹ ID, MS; Nancy Robles¹ ID, NP-C; Lisa Von Wald, APRN; Wendy Flowers, PhD; Gregory L. Hershberger, MS; Krishna G. Abraham¹ ID, MD, PhD; Michael A. Burke¹ ID, MD; Jamie Diamond¹ ID, MD; Mark H. Drazner¹ ID, MD; Gregory A. Ewald¹ ID, MD; Stephen S. Gottlieb¹ ID, MD; Garrie Haas¹ ID, MD; Mark Hofmeyer¹ ID, MD; Gordon S. Huggins¹ ID, MD; Javier Jimenez¹ ID, MD, PhD; Daniel P. Judge¹ ID, MD; Stuart Katz¹ ID, MD; Masataka Kawana¹ ID, MD; Evan P. Kransdorf¹ ID, MD, PhD; Cindy M. Martin¹ ID, MD; Elina Minami, MD; Anjali Owens¹ ID, MD; Palak Shah¹ ID, MD; Chetan Shenoy¹ ID, MD; Supriya Shore¹ ID, MD; Frank Smart¹ ID, MD; Douglas Stoller¹ ID, MD, PhD; Jose Tallaj¹ ID, MD; W.H. Wilson Tang¹ ID, MD; Jessica Wang¹ ID, MD; Jane Wilcox, MD; Ray E. Hershberger¹ ID, MD

BACKGROUND: Clinical genetic evaluation for patients with dilated cardiomyopathy (DCM) is minimally implemented, and models of care are not well defined. To understand current genetic care for DCM, a systematic needs assessment was conducted.

METHODS: Principal investigators of the DCM Consortium convened at the Summer Scientific Symposium in July 2025. An electronic needs assessment was conducted among the 24 principal investigators in advance to define current care models by evaluating which genetic evaluation components recommended by the Heart Failure Society of America were conducted, by whom, and the time required for each component. Descriptive statistics were generated to characterize model features. Focus group discussions explored barriers and facilitators to implementing genetic services.

RESULTS: Four care models emerged from the principal investigator responses: model 1: Traditional-Synchronous (25%, n=6, requiring the most time per patient); model 2: Traditional-Asynchronous (33%, n=8); model 3: Externally Sourced (17%, n=4); and model 4: Physician/Advanced Practice Provider Conducted (25%, n=6, requiring the least time per patient). All models used genetic testing, whereas other components were implemented variably or not at all. Models 1 (15.7±4.1) and 2 (15.4±3.0) were rated more acceptable than model 4 (9.8±2.9; model 1 versus model 4; $P=0.027$; model 2 versus model 4; $P=0.023$). Notably, 88% of principal investigators used genetic information for treatment decisions, including implantable cardioverter defibrillator placement (83%; n=20) and cardiac transplantation (63%; n=15). Major facilitator themes from focus group discussions included having a genetic counselor as part of the heart failure team and developing authoritative standards directing provision of DCM genetic services. Barrier themes included operational challenges, limited personnel, clinician under-recognition, need for new service delivery models, and billing/reimbursement.

CONCLUSIONS: DCM genetic care models and components were highly variable across the 24 sites of the DCM Consortium, although all sites discussed similar factors that enable or hinder the implementation of genetic services for DCM. Understanding the basis of practice model variability may provide insight to yield more scalable care approaches.

Key Words: cardiomyopathy, dilated ■ genetic testing ■ heart failure

Correspondence to: Elizabeth Jordan, MS, CGC, Division of Human Genetics, Department of Internal Medicine, The Ohio State University Wexner Medical Center, Biomedical Research Tower, Room 306, 460 W 12th Ave, Columbus, OH 43210, Email elizabeth.jordan@osumc.edu; or Ray E. Hershberger, MD, Division of Cardiology, Department of Internal Medicine, The Ohio State University Wexner Medical Center, Biomedical Research Tower, Room 304, 460 W 12th Ave, Columbus, OH 43210, Email ray.hershberger@osumc.edu

This manuscript was sent to Onyedika J. Ilonze, MD, MPH, Guest Editor, for review by expert referees, editorial decision, and final disposition.

Supplemental Material is available at <https://www.ahajournals.org/doi/suppl/10.1161/CIRCHEARTFAILURE.126.014544>.

© 2026 American Heart Association, Inc.

Circulation: Heart Failure is available at www.ahajournals.org/journal/circheartfailure

WHAT IS NEW?

- Four care models to deliver genetic evaluation services for the assessment of dilated cardiomyopathy (DCM) were identified among the 24 sites of the DCM Consortium.
- Despite universal use of genetic testing and shared barriers and facilitators across sites, substantial variability exists across and within the identified model types, and clinical genetic evaluation is not routinely implemented in all patients with DCM.

WHAT ARE THE CLINICAL IMPLICATIONS?

- For heart failure centers seeking to implement genetic services for patients and families with DCM, the DCM Consortium describes the scaffolding of current care models implemented at heart failure centers across the United States to inform local program development.
- The incomplete and variable implementation of care models highlights the need for a standardized, scalable framework to enhance consistent and equitable integration of clinical genetic evaluation into routine DCM care.

Nonstandard Abbreviations and Acronyms

APP	advanced practice provider
DCM	dilated cardiomyopathy
GC	genetic counselor
GT	genetic testing
HCM	hypertrophic cardiomyopathy
HF	heart failure
PI	principal investigator

Dilated cardiomyopathy (DCM), a major cause of heart failure (HF), has a substantial genetic basis.¹ Current scientific statements^{2,3} and guidelines⁴⁻⁶ recommend genetic evaluation for patients with DCM; however, implementation has been challenging, and uptake of genetic testing (GT) has been minimal.⁷⁻⁹ Others have addressed the underutilization of clinical genetics practice in cardiomyopathy,⁷⁻¹⁰ and investigations of genetic services for all heritable cardiovascular diseases have demonstrated similar challenges, with barriers including education, cost, and lack of access to genetic counselors (GCs) and clinical genetics expertise within the care team.^{11,12}

In 2024, the investigators of the DCM Consortium,¹³ a group organized to conduct genetic research in DCM with previous research contributions,^{14,15} convened to identify issues and target efforts to improve the clinical implementation of precision medicine for DCM in their US HF clinics. The results of that needs assessment identified major challenges related to billing, workforce,

education, and management of genetic uncertainty.¹⁶ To gain further insight into these issues and to consider remedies, in 2025, the DCM Consortium investigators sought to extend their prior work with a systematic investigation of each institution's current clinical genetics care model for patients with DCM at their centers and to codify each institution's operational approach. Here, investigators of the DCM Consortium provide a survey of the characteristics, strengths, and challenges of their current models and provide a blueprint for HF clinicians seeking to implement genetic services at their institutions.

METHODS

Data Availability

The single institutional review board for the DCM Research Project at the University of Pennsylvania deemed this assessment exempt. The survey instrument used in this study and the final codebook of transcripts of interviews have been provided in the [Supplemental Material](#). Additional data from this study can be made available to interested investigators on reasonable request.



Survey Design

A web-based Qualtrics survey was completed by site principal investigators (PIs, all cardiologists) of the DCM Consortium between June 20 and July 11, 2025. There were 4 sections to the survey: (A) Your/your institution's information, (B) Genetic evaluation practices, (C) Operational approach to genetic evaluation, and (D) Goals for the future and final comments. Section C (Operational approach to genetic evaluation) included the acceptability implementation measure,¹⁷ in which participants rated 4 statements about their current practice model on a Likert scale from 1 (completely disagree) to 5 (completely agree) to the acceptability of their current model, where acceptability is defined as the perception among implementation stakeholders that a given treatment, service, practice, or innovation is agreeable, palatable or satisfactory.¹⁸ The acceptability implementation measure has been validated previously, demonstrating good internal consistency (Cronbach $\alpha=0.83$) and strong test-retest reliability (attenuation-corrected Pearson correlation=0.80).¹⁷ The full survey instrument is provided in the [Supplemental Material](#).

Focus Group Discussion

Open text responses and section C items (Operational approach to genetic evaluation) were analyzed to identify patterns in practice structures and workflows and to group similar care approaches in advance of the in-person 2025 Summer Scientific Symposium event. Two investigators (E.J. and T.M.) coded the responses independently to identify themes at each center that distinguished similar patterns and defined model types. Focus group discussions with DCM Consortium site PIs and other site team representatives present at the Symposium were conducted within each group, as identified and assigned by the investigators. A structured question guide was used to guide the discussion groups, which were facilitated by a site-affiliated GC present at the Symposium. The Consolidated

Criteria for Reporting Qualitative Research checklist¹⁹ and coding tree are provided in [Tables S1 and S2](#).

Analysis

Descriptive statistics (percentages, medians, means, and SDs) were calculated. Acceptability implementation measure scores were calculated by summing the 4 Likert-scale item scores (range, 4–20) consistent with the validated scoring approach.¹⁷ Because the responses were ordinal and not normally distributed, group differences were assessed using Kruskal-Wallis tests, followed by Dunn post hoc tests with Šidák correction for pairwise comparisons. Statistical significance was defined as $P<0.05$.

Focus group discussions were recorded, transcribed, and analyzed for themes. As defined by the Consolidated Framework for Implementation Research, barriers were considered factors that hinder implementation of DCM genetic services, and facilitators were considered factors that enable or support successful integration into care.²⁰ Transcripts were coded for themes using the constant comparative method,²¹ a reductive and interpretive method that allows themes representing meaningful patterns of common messages to emerge. Two investigators independently read and analyzed the transcribed data (T.M. and H.K.), resulting in 2 independent sets of codes. The investigators generated independent notes with tentative interpretations of explicit and implicit meanings and comparisons of the content with other sources. The investigators met to compare notes and refine the codebook. The investigators (T.M. and H.K.) independently coded each transcript and then met to adjudicate passages and codes. All 4 transcripts were coded and recoded using the updated codebook. The final codebook and code counts are provided in [Tables S2 and S3](#).

RESULTS

All site PIs (n=24, 100%) participated in the survey. Demographic and training backgrounds of the 24 site PIs have been published previously: most were non-Hispanic White males who were HF/transplant-trained cardiologists based at academic medical centers across the United States ([Figure S1](#)). Site characteristics included active HF/transplant inpatient and outpatient programs, ranging in size from medium to large volumes of patients and selected to maximize diversity of patient race and ethnicity, as previously presented.^{13,16}

Care models are defined in [Figure 1](#) and described below. Clinical team members participating in each care model are summarized in [Table 1](#), with an average of 5.6 HF/transplant physicians per site participating in genetic evaluation. Seventeen (71%) centers had access to GCs within their institution, including 6 with cardiovascular-specialized GCs integrated into the HF or cardiology team. One site had access to genetics expertise, but not within their institution, and 6 did not have access to any genetics professionals. On average, 1.6 advanced practice providers (APPs) also participated in genetic evaluation component(s), with 29% of centers having an APP with genetics expertise on their team.

Four Care Models for DCM Genetic Evaluation

Thematic analysis of survey responses revealed 4 models of care, as defined in [Figure 1](#): model 1: Traditional-Synchronous (n=6); model 2: Traditional-Asynchronous

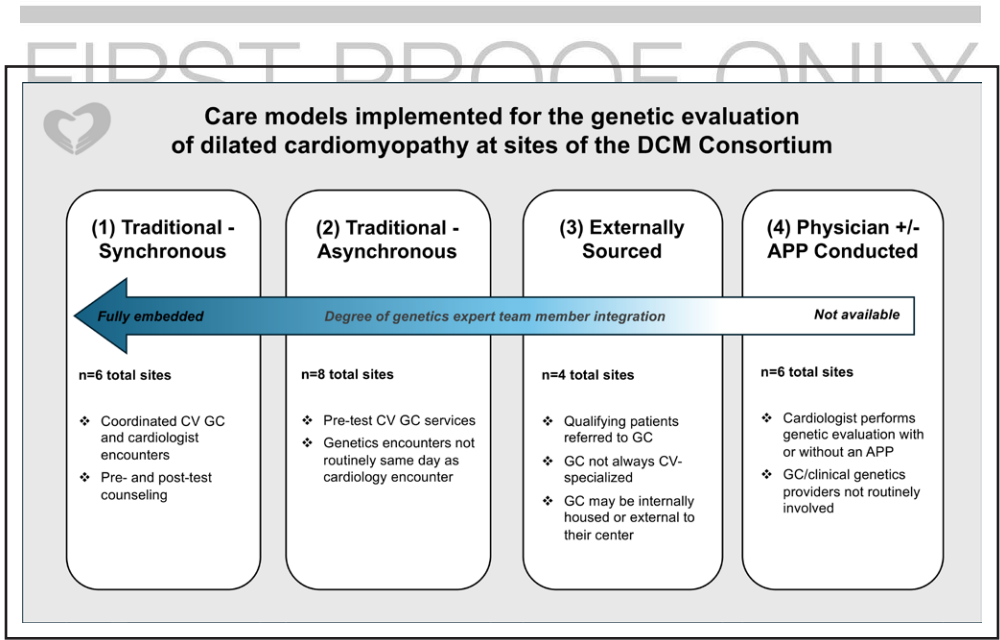


Figure 1. Current care models for genetic evaluation of dilated cardiomyopathy at sites of the Dilated Cardiomyopathy (DCM) Consortium.

The arrow points to increasing degrees of genetics expert team member integration into the model of care, with darkened color showing higher integration; expert team members include genetic counselors and geneticists. APP indicates advanced practice provider, including a certified nurse specialist or nurse practitioner; CV, cardiovascular; DCM, dilated cardiomyopathy; GC, genetic counselor; and pretest, before genetic testing.

Table 1. Physician and Advanced Practice Provider Participation in the DCM Genetic Evaluation by Practice Model

	All sites (N=24)	Model 1 (n=6)	Model 2 (n=8)	Model 3 (n=4)	Model 4 (n=6)
How many HF/TX cardiologists participate in any aspect of a DCM genetic evaluation? mean (median)	5.6 (4)	3.8 (2.5)	6.6 (8)	8 (4)	4.5 (3)
A genetic counselor internal to their institution selected is the primary team member conducting any component of a DCM genetic evaluation. Yes, n (%)	17 (71%)	6 (100%)	8 (100%)	3 (75%)	0 (0%)
Do you have >1 APP who has specific genetic expertise or experience in cardiomyopathy at your program? Yes, n (%)	7 (29%)	2 (33%)	3 (37.5%)	1 (25%)	1 (16.7%)
How many APPs in your HF/TX program participate in any aspect of a DCM genetic evaluation? mean (median)	1.6 (1)	0.83 (0)	2.25 (1.5)	1.75 (1.5)	1.67 (1)
Do you have access to a clinical research coordinator or other dedicated research staff who works with your patient population with DCM for genetics-related research? Yes, n (%)	10 (42%)	5 (21%)	3 (13%)	0 (0%)	2 (8%)

Aspects of the genetic evaluation were defined as “ordering testing, genetic education, conducting informed consent, returning results, coordinating family follow-up.” Model 1=Traditional-Synchronous; Model 2=Traditional-Asynchronous; Model 3=Externally Sourced; and Model 4=Physician/APP Conducted. APP indicates advanced practice provider, defined as a certified nurse specialist or a nurse practitioner; DCM, dilated cardiomyopathy; and HF/TX, heart failure/heart transplant.

(n=8); model 3: Externally Sourced (n=4); and model 4: Physician±APP Conducted (n=6). Estimated time required to implement genetic evaluation was assessed using predefined categorical options. From these responses, the estimated time required to implement the core components (Figure 2), as defined by the Heart Failure Society of America,⁵ varied widely for each model, with model 1 requiring the most time per patient (128 minutes) and model 4 requiring the least (67.9 minutes; Table 2). Composite scores of acceptability using the acceptability implementation measure differed across model types, where model 1 (15.7±4.1) and model 2 (15.4±3.0) were overall rated significantly

more acceptable than model 4 (9.8±2.9; model 1 versus model 4; *P*=0.027; model 2 versus model 4; *P*=0.023; Table 3). Item-level pairwise comparisons between model types are provided in Table S4.

Model 1: The Traditional-Synchronous Model

Six sites used a Traditional-Synchronous model. The common theme defining this model included a cardiovascular-specialized GC and cardiologist (synchronous) encounters coordinated within a single, multidisciplinary clinic visit for patients. All model 1 sites provided pre-GT counseling or education and family history assessment (Figure 2), with support from an embedded

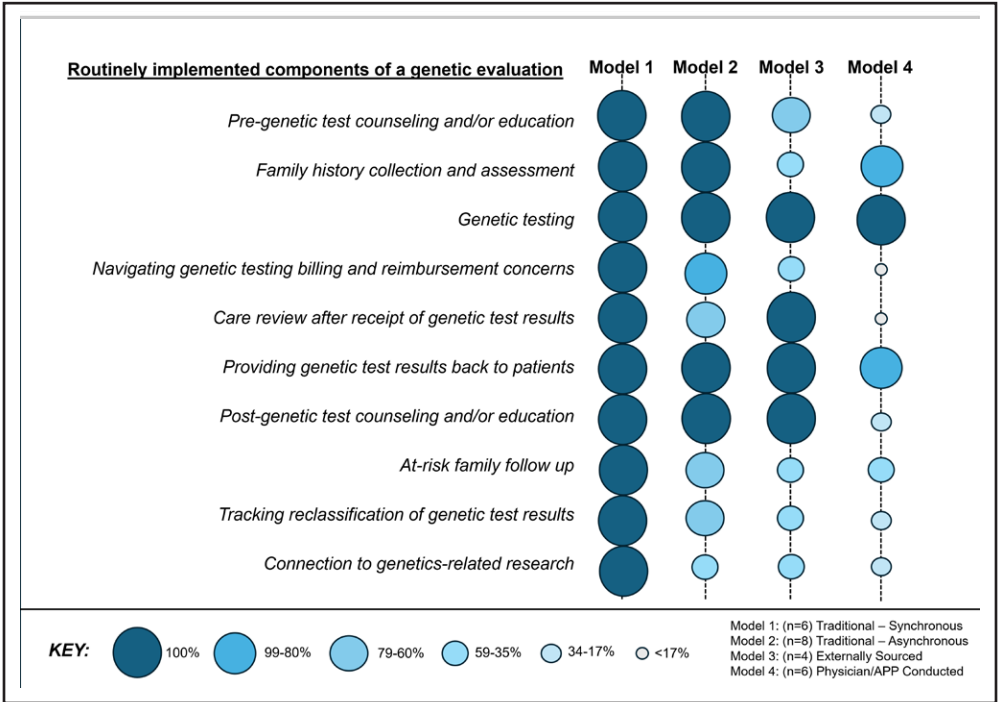


Figure 2. Aspects of a genetic evaluation routinely offered in the currently implemented models for patients with dilated cardiomyopathy (DCM) at DCM Consortium sites.

The percentage of sites that routinely implement components of a genetic evaluation, specified by models of genetic care, is shown by the size and color of the circles. APP indicates advanced practice provider, including a certified nurse specialist or nurse practitioner.

Table 2. Time Required to Implement Components of a DCM Genetic Evaluation for a Single Patient With DCM Under Each Model Type at DCM Consortium Sites

Model type	Estimated time (min) to complete evaluation components for an individual patient with DCM					
	Pre-genetic test counseling*	Family history assessment	Navigating billing and reimbursement*	Result review	Post-genetic test counseling*	Total time to complete all components†
Model 1: Traditional-Synchronous (n=6)	35.5	12.5	21.8	34.3	24.3	128.4
Model 2: Traditional-Asynchronous (n=8)	39.9	9	16.8†	21.8†	24.1†	111.5
Model 3: Externally Sourced (n=4)	23†	11†	15.5†	30.5	15.5	95.5
Model 4: Physician±APP Conducted (n=6)	15.5†	6.4	N/A	23†	23†	67.9

N/A denotes not applicable because no sites in this model group indicated that they navigate billing and reimbursement for genetic services for patients with DCM on a routine basis. APP indicates advanced practice provider; DCM, dilated cardiomyopathy; and min, minute(s).

*Responses included options of ≤15, 16–30, 31–60, or >60 min; medians for these structured ranges were used to calculate average estimated time for each of these items, respectively.

†Not all sites in this model group indicated that they routinely perform this service component; therefore, the denominator is lower than the total sites in the model group.

‡Total time was calculated as the cumulative average time for each component represented across the table for each respective model.

GC (100%, n=6) in addition to an APP (16.7%, n=1) or, in some cases, provided by the cardiologist themselves (33%, n=2). Only 1 site offered genetic evaluation services in-person only, whereas others offered both telehealth and in-person options (n=5). GCs were the primary team members who returned results to patients, using phone call disclosures across all centers but also offering in-person disclosures (50%, n=3), electronic health record messaging (66%, n=4), and mailed letters (16.7%, n=1). Post-GT GC encounters were provided for patients with all types of results (positive, negative, uncertain, and incidental findings). Model 1 was the most time-intensive, requiring an estimated cumulative average of 128 minutes per patient to complete all key components (Table 2). This model also provided family-based support post-GT.

Model 2: The Traditional-Asynchronous Model

The Traditional-Asynchronous model was the most frequently implemented across DCM Consortium sites (n=8), in which the cardiology and genetics encounters

were conducted separately (asynchronously) during 2 separate clinic appointments while still following a historically typical workflow, including genetic counseling both before and after GT (Figure 1). Like model 1, GCs were identified as primary team members providing pre-GT counseling and education at the majority of sites (n=7). Four sites (50%) also listed a cardiologist (including themselves) as a team member providing aspects of this service in addition to GCs. A cardiovascular-specialized GC internal to the institution was a primary team member at all sites that provided results back to patients (100%, n=8), in addition to the cardiologist at 2 centers (25%) and an APP at 1 (12.5%). There were several models of post-GT result communication, with most model 2 sites offering multiple approaches, including telephone (n=5), in-person (n=5), electronic health record message/letter (n=5), and mailed letter (12.5%, n=1). Model 2 sites provided genetic evaluation components variably (Figure 2), including billing/reimbursement support (87.5%, n=7), reviewing care implications upon receipt of genetic results with the clinical team

Table 3. Acceptability Implementation Measure of Current Models of Genetic Evaluation for DCM at Sites of the DCM Consortium

Item	Model 1 (M1; n=6)	Model 2 (M2; n=8)	Model 3 (M3; n=4)	Model 4 (M4; n=6)	Significant pairwise differences*
	Mean±SD				
My institution's current practice model for the genetic evaluation of DCM meets my approval	3.8±1.2	3.9±0.8	2.3±1.3	2.8±0.9	M2 vs M3 (<i>P</i> =0.038)
My institution's current practice model for the genetic evaluation of DCM is appealing to me	4.2±1.1	3.9±0.8	2.5±1.7	2.5±0.8	M2 vs M3 (<i>P</i> =0.038)
I like my institution's current practice model for the genetic evaluation of DCM	3.8±1.1	3.9±0.8	2.3±1.3	2.3±0.7	M1 vs M4 (<i>P</i> =0.026), M2 vs M3 (<i>P</i> =0.040), M2 vs M4 (<i>P</i> =0.013)
I welcome my institution's current practice model for the genetic evaluation of DCM	3.8±1.1	3.8±0.8	2.3±1.6	2.2±0.7	M1 vs M4 (<i>P</i> =0.028), M2 vs M4 (<i>P</i> =0.026)
Composite score	15.7±4.1	15.4±3.0	9.25±5.8	9.8±2.9	M1 vs M4 (<i>P</i> =0.027), M2 vs M4 (<i>P</i> =0.023)

Model 1=Traditional-Synchronous; model 2=Traditional-Asynchronous; model 3=Externally Sourced; and model 4=Physician/APP Conducted. Means±SD are shown for descriptive purposes. Statistical differences across groups were assessed using Kruskal-Wallis tests, followed by Dunn post hoc tests with Šidák correction for pairwise comparisons. Significance was set at $\alpha=0.05$. Only significant comparisons ($P<0.05$) are shown. DCM indicates dilated cardiomyopathy.

*All pairwise comparisons and corresponding P values at the item level and composite scores are provided in Table S4.

(75%, n=6), family follow-up (75%, n=6), reclassification tracking (75%, n=6), and connection with research (50%, n=4).

Model 3: Externally Sourced Genetics Services

The central feature of the model 3 sites (n=4) was referral of patients to genetics providers in a department, institution, or company external to the HF or cardiology team for coordination of GT (Figure 1). Beyond these components, who, how, and when other aspects of the genetic evaluation were conducted was variable. One model 3 site had an APP with specific genetic expertise or practice experience in cardiomyopathy to support the program (Table 1), but it was noted that this individual was primarily assigned to hypertrophic cardiomyopathy (HCM) patient navigation. Only 2 of the model 3 sites routinely collected and assessed family history as part of the genetic evaluation service. Test results were typically communicated back to the patient by the externally referred genetics provider (50%), sometimes in addition to the cardiologist sharing the results themselves (50%) through multiple mechanisms, including telephone (100%), electronic health record communication (75%), or in-person (50%). One model 3 site only offered post-GT counseling for positive results (pathogenic or likely pathogenic variants), whereas post-GT counseling was provided for all result types at the other model 3 sites. Half of the model 3 sites provided billing/reimbursement support, reclassification tracking, and connections with research, and all reported that they conducted care reviews after receipt of genetic results.

Model 4: Physician/APP Conducted Model

Model 4 (n=6 sites) was characterized by a cardiologist with or without the support of an APP partner conducting the genetic evaluation. This was the least time-intensive model, with a cumulative estimate of about 68 minutes required per patient to perform all implemented

components (Table 3). Of the model 4 sites that provided pre-GT services (n=2), both reported that only patient education was provided, and this could be done by any clinical team member on the same day the patient was identified in clinic. Results were provided back to patients by various team members, including the cardiologist (66.7%, n=4), a registered nurse (33%, n=2), an APP (16.7%, n=1), or a medical assistant (16.7%, n=1); 1 site did not specify who disclosed the results.

Genetic Testing Practices

Although implementation of the Heart Failure Society of America–recommended components of the genetic evaluation varied widely across and within model groups, all sites across model types indicated that they routinely conducted GT for DCM in their model (Figure 2). The sample types and test selection varied widely across centers and within practice models (Table 4). Twenty-two PIs stated that GT results changed care for varying proportions of the patients they evaluated: in 1-5% of patients (n=2 PIs), 5-10% (n=6), 10-20% (n=9), 20-50% (n=4), or 50-100% (n=1). Only 2 site PIs stated that genetic information never or rarely changed care. When asked if genetic information had been used to make treatment decisions for at least 1 patient with DCM, 87.5% (n=21) selected at least 1 treatment decision, with implantable cardioverter defibrillator implantation (83%; n=20) and cardiac transplantation (62.5%; n=15) being the most common (Figure 3). Respondents perceive the ability to do GT for HCM as easier to accomplish than for DCM at 42% (n=14) of centers.

Billing and Reimbursement Practices

GT was billed to insurance at more than half (54%, n=13) of the sites. Others used sponsored GT programs (n=1) or were not sure or did not specify billing practices

Table 4. Sample Collection and Genetic Testing Strategies at DCM Consortium Sites by Model Type

	Model 1 n=6 (%), (n)	Model 2 n=8 (%), (n)	Model 3 n=4 (%), (n)	Model 4 n=6 (%), (n)
Sample type collected				
Blood	17%, 1	0	25%, 1	17%, 1
Saliva	0	25%, 2	0	0
Buccal	33%, 2	25%, 2	25%, 1	33%, 2
Blood or saliva	17%, 1	0	25%, 1	0
Blood or buccal	0	38%, 3	0	0
Blood, saliva, or buccal	33%, 2	12%, 1	25%, 1	50%, 3
Panel ordered				
DCM targeted	0	12%, 1	25%, 1	17%, 1
Comprehensive cardiomyopathy	33%, 2	50%, 4	25%, 1	66%, 4
Comprehensive cardiomyopathy and arrhythmia	66%, 4	38%, 3	50%, 2	17%, 1

Percentages may not add to 100 due to rounding. DCM indicates dilated cardiomyopathy.

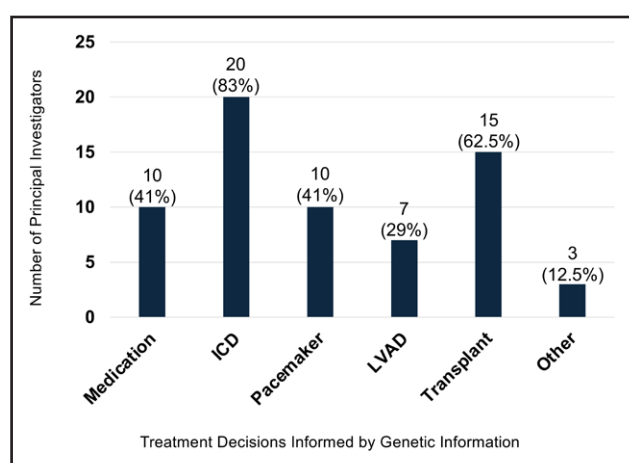


Figure 3. Treatment decisions informed by genetic information for at least 1 patient with DCM by cardiologist site principal investigators of the DCM Consortium.

Other selections included exercise restrictions (1), screening for sudden death with cardiac monitoring (1), and frequency of follow-up (1). ICD indicates implantable cardioverter defibrillator; and LVAD, left ventricular assist device.

(n=10). Of the 17 sites (71%) that indicated that they routinely supported patients in navigating billing and reimbursement concerns, a majority (82%, n=14) stated that they did not know how often insurance denials for GT were received; 1 site did not specify, and those that did estimate (n=2) projected that 5-10% of patients experienced denials. Thirteen PIs stated that they did not know how often their patients were responsible for an out-of-pocket cost higher than desired. Of the 3 sites that provided estimates, it was projected that patients would receive high out-of-pocket costs 10% (n=1) or 20% (n=2) of the time. In contrast, the estimated frequency of patients expressing concern about the cost of GT ranged widely across centers, from 5% to 75% (mean=27.5%; median=20%) for those providing an estimate (n=10). On a 5-point scale of 1 (not at all concerned) to 5 (very concerned), PIs had generally neutral levels of concern (mean=2.8) related to the cost of testing. Regarding billing for GC services, 83% responded yes and 17% responded maybe; improved reimbursement for GCs would improve the ability to add a GC to the HF team.

Focus Group Discussion

Themes related to facilitators, or enabling factors, and barriers, or hindering factors,²⁰ to genetic evaluation emerged from within-model small group discussions. These themes and exemplary quotes are shown in Table 5. Having access to a GC on the HF clinical team was a consistently named facilitator to navigating the genetic evaluation process. All groups also discussed the supportive impact that authoritative recommendations directing routine implementation of genetic services in

DCM could have, creating a foundation for a best practice accreditation framework for DCM care, similar to the current center of excellence designation available for HCM care, where a genetic evaluation would be a required service available to all qualifying patients with DCM to meet care standards for qualifying HF centers.

Internal institutional barriers and external factors hindering implementation of DCM genetic services that were discussed included operational challenges, limited personnel resources, including an insufficient GC workforce, and an adverse cost and billing landscape with poor reimbursement for genetic services. Lack of scalability of historical models, termed *Traditional* herein, was also discussed as a barrier, with participants sharing successes with how they have adapted traditional processes for improved efficiency. Model groups also reflected that there was under-recognition of the genetic contributions to DCM among their HF colleagues, which may increase the routine use of genetic services in clinical care if better understood.

DISCUSSION



We have provided a comprehensive assessment of how genetic evaluations for DCM were conducted at 24 academic medical centers, all of which were engaged in DCM genetic research and were committed to clinical genetic care for patients with DCM and their families. Using a systematic questionnaire and in-person focus group interviews as a follow-up to a prior effort,¹⁶ we identified 4 different practice models for genetic care for patients with DCM. Although GT was a universal component of all models, other model characteristics varied greatly, as did the time, effort, and expertise of the personnel needed to conduct genetic evaluations.

This assessment of care models has provided practical insights for others developing cardiomyopathy genetic evaluation programs and seeking to elevate their clinical service beyond “just the test.” Genetic evaluation of cardiomyopathy is informed by practice guidelines from the Heart Failure Society of America⁵ and the American College of Medical Genetics and Genomics,³⁻⁶ which define a genetic evaluation as “a systematic approach that includes a comprehensive family history, phenotypic evaluation of the proband and at-risk family members, genetic counseling, genetic testing, if indicated, with pre- and posttest counseling, and guidance as indicated for specific drug and/or device or other specific therapeutic interventions.” We emphasize that GT is a minimum necessary component but alone does not define the broader service of a genetic evaluation, which extends the clinical impact of genetic information to patient care and family risk management. Until genetic evaluations for DCM are able to be implemented routinely for all qualifying patients, patients and families

Table 5. Themes That Emerged From Focus Group Discussions Within Model Type Groups

Theme	Subtheme	Demonstrative quotes
Facilitators	Genetic counselors as members of the HF team	It's just a relief, I think, to have ... genetics Most of our colleagues are not at all experts in the genetics side of things. So being able to have an expert in genetics, also an advantage. –MD, Traditional-Asynchronous Model
		Genetic counseling works. Well, patients are satisfied, and they'll travel long distances to see someone who's got the rare condition and understands the gene that they're troubled with, and a positive result that's involving their family. And so there's a lot of positive sides there. –MD, Traditional-Synchronous Model
		The other problem is ... the variants in genes ... the what to do about it, you know, and how it relates specifically to this [patient's] phenotypes. I think it's difficult, at least for me, because I don't have enough knowledge to really understand the little nuances of ... if that is going to change my management ... And that's why, again, today, genetic counseling experience, that discussion with that individual, may help me make that decision which I don't have right now... –MD, Physician/APP Conducted Model
	Authoritative guidance directing genetic services as standard of care	And HCM [hypertrophic cardiomyopathy] clinic actually is a little bit streamlined [for] new referrals for HCM. At our center [referrals] are almost always coming from outside the system. And the director for HCM [clinic] actually has it set up that now [a] genetics counselor visit, upfront for pretest is part of the streamlined package of stuff that you get when you come ... –MD, Traditional-Asynchronous Model
		They did something like that in the PAH [pulmonary arterial hypertension] administration ... trying to create PAH centers of excellence ... It's kind of like the idea of like cardiovascular genetic center of excellence, right? ... And that was something, because there was a roadmap, and I don't think we have a roadmap. –MD, Physician/APP Conducted Model
		I know our institution ... is more willing to pay for genetic counselors or administrative assistants, if it's part of that "Center of Excellence" tag. –GC, Physician/APP Conducted Model
Barriers	Operational challenges to genetic testing	There is a drop [off] from wanting to do the test to actually getting tested ... [it's a] big problem. There are people that will do [a saliva sample] like two or three times ... I think a lot of it is just the systems of testing is just not smooth? –MD, Externally Sourced Model
		It's hard to even find a variant [in the medical record]... We actually have Epic Genomics ... We've got a package in Epic, but we don't even use it because it takes triple the amount of time to put [in] the information ... Actually I had a slideshow of what are the 10 ways [you] can find [a genetic test result in Epic]. Five of them is completely invisible, but really kind of the tracking solutions and downstream is the biggest. –MD, Externally Sourced Model
	Under-recognition of the genetic causes of DCM	But basically, the doctors just don't see that 30% of the patients actually may have a genetic basis [for their DCM]. That's the problem, that lack of knowledge. When you put it in that perspective, then you would pay attention. But I think it's something that has not sunk in there ... And also the other thing is what to do. This is also the other part, you know also. So then what to do with that? It's not really part of the pathway that I think most cardiologists have in their mind. –MD, Physician/APP Conducted Model
		I've made a lot of way and a lot of education to try to get other cardiologists to do that [genetic testing] independently of me. It actually has worked ... because many times I get someone asking me, "Well, how do I do that?" Or "What do I do with this result?" They don't have a structured genetic counseling team in cardiovascular disease which I've been asking for and trying to [get]. –MD, Physician/APP Conducted Model
	Limited personnel resources	We have 11 advanced heart failure cardiologists, but only several of us are using [the] genetic counselor. And there are real scalability issues. That if we were to, if the entire group started doing it, I think we'd swamp the resources we have. –MD, Traditional-Asynchronous Model
		We in fact need more administrative support. Hiring a GC doesn't take care of the problems. We can't accommodate volumes. They don't have infrastructure in place to support those. –MD, Traditional-Asynchronous Model
		I think the biggest challenge is volume and I think balancing the benefits of pretest counseling with just getting people tested. –GC, Externally Sourced Model
	Need for new service delivery strategies	Having the split model [MD + GC clinic and separate GC only clinic], has helped a lot with wait times. That was one of the biggest issues we identified was just that it was getting filled up, and people were waiting a long time a lot of time. –GC, Traditional-Synchronous Model
		If someone's already seen one of my ... heart failure colleagues, I don't need to see them, but a GC needs to see them. And so those are done separately, usually much faster and much more efficiently. –MD, Traditional-Synchronous Model
		We do a lot of inpatient counseling. But I think it [inpatient counseling] really catches a ... more diverse population ... Inpatient time is when family members are around. Then, you know, we can actually talk about family screening at the same time ... So we definitely try to encourage all the heart failure faculty members, or even any cardiologist who's attending in the hospital. We encourage them to just refer inpatient. –MD, Traditional-Synchronous Model
	Cost, billing, and reimbursement	The companies are evolving, the reimbursement is changing. It's hard even for the genetic counselor to keep up sometimes. –MD, Traditional-Asynchronous Model
		Sometimes I end up seeing [the patient] not because the actual need for seeing the patient [but for billing]. There's a lot of Medicare patients. –MD, Traditional-Synchronous Model
		It comes down to money. We reorganized our system into a service line model where, instead of the money for procedures, etc, that are done in the hospital, going to the hospital ... Everything in cardiology ... comes into one pot, and suddenly they have the money to hire a GC. –MD, Traditional-Asynchronous Model
		... insurance [companies] think of ... 1 year term ... budget and expenses. So they don't really care about what happens [in] 5 y. That's why they don't really encourage ... preventive medicine, [or] to invest in that, because they only care about the year." –MD, Physician/APP Conducted Model

DCM indicates dilated cardiomyopathy; and HF, heart failure.

will suffer from the missed opportunities that precision care for DCM now offers.¹

Although all models consistently implemented the minimum component of GT, the key differentiating characteristic across models centered on whether a clinical genetics team member, typically a GC but in some cases an APP with a clinical genetics focus, was available to partner with or support physicians with genetic evaluation components, and, if so, whether they were housed internally or externally to the HF team or institution. Models 1 and 2 used a traditional model inclusive of pre- and posttest genetic counseling services provided by cardiovascular-specialized GCs. This workflow has been historically considered a gold standard approach and rated as significantly more acceptable¹⁷ in the current report. Access to a GC has repeatedly been identified as a facilitator to implementing clinical genetics services for cardiovascular disease.¹⁶ Although the historically embedded model (termed here as a Model 1: Traditional-Synchronous) provided comprehensive services, such a standard may compromise a more equitable provision of services due to the lack of scalability. Notably, the traditional model was the most time- and resource-intensive approach, requiring nearly double the time compared with the other models reported herein. Alternative models of service delivery are actively being piloted internationally in multiple specialty areas, including HF clinics.^{22,23} To fully realize the potential of new approaches, improved billing and reimbursement procedures are needed in the United States, which has been a consistent major barrier to genetics services.^{2,24} New billing codes for GC services are available, and the community has continued to appeal to Centers for Medicare and Medicaid Services²⁴ to acknowledge GCs as billable providers.^{3,5,25–27} Although lack of GC support was a barrier theme emerging from the discussions of all model groups, GC-integrated programs were found to generate a positive downstream revenue,²⁸ underscoring the fiscal value of building genetics expertise into cardiovascular programs.

The broad-based discussions brought forward additional insights relevant to considering the implementation of genetic care. In an open text item, a PI suggested that cardiologists need to “lower the bar to refer DCM patients for genetic evaluation,” a sentiment that aligns with the barrier of under-recognition of genetic contributions to DCM among the HF clinical community (Table 5). Indeed, all qualifying patients with DCM and their family members, regardless of the severity of their personal or family history at diagnosis, should have the opportunity to seek precision care guided by the remarkable and emerging benefits that a genetic diagnosis of cardiomyopathy can now facilitate. However, as we lower the bar to ensure that all patients with DCM have access, we must also take care to raise the bar for clinical care standards as novel, scalable adaptations to current models are developed.

It was also speculated in an open-response item that “The oncology and obstetrics models of genetic evaluations are much further ahead because genomic evaluation is part of their clinical management processes and not taken out of the practice pathway.” Indeed, there are authoritative and time-specific recommendations for GT and genetic counseling from organizations such as the American College of Obstetrics and Gynecology^{29–32} and the National Comprehensive Cancer Network,³³ driven by scientific advancements that established a direct impact of genetic information on immediate treatment and care decisions in these areas. Underlying the remarkable uptake of cancer genetics is the direct impact of somatic genetic signatures in cancer tissue on chemotherapeutic treatment decisions. Within the DCM and HF care paradigm, the evidence base for germline genetic data has not been established to directly inform treatment algorithms. Such contemporaneous, rigorous, and well-designed studies are needed to develop risk models for patients with HF and DCM that include genetic information to inform treatment decisions, as underscored by the results reported here that ≈88% of PIs, all of whom are HF/transplant cardiologists, already use genetic information to inform clinical decisions based on conventional wisdom, expert consensus, and clinical experience.

The highly variable models of genetic care shown in this study suggest that a more sustained and substantial evaluation coupled with innovation is needed for DCM care models, as DCM is the most common cardiomyopathy and accounts for a sizable proportion of HF with reduced ejection fraction. Patients with HF with reduced ejection fraction now commonly receive care within HF clinics. Interestingly, 42% of DCM Consortium site PIs reported that genetic evaluations are more routinely accomplished for patients with HCM than for patients with DCM. Notably, HCM clinics at centers meeting an HCM Center of Excellence designation require genetic services. In contrast, defining the genetic cause of DCM in HF clinics has not been a clinical priority.³⁴ We suggest one insight derived from this analysis is that developing and establishing an authoritative best practice framework that includes clinical genetics practice guidance and accreditation mechanisms for the care of patients with all cardiomyopathies is needed. If achieved, such a framework may drive awareness and implementation of genetic medicine for all patients with HF and DCM and other related cardiomyopathies (eg, arrhythmogenic right ventricular cardiomyopathy and HCM), as applicable, who are seen in HF or other relevant clinics.

Limitations

This assessment represents only the experiences and perspectives of those of the DCM Consortium, all of whom are members of advanced HF/transplant programs at academic medical centers and therefore is not

representative of practices beyond academia or of all HF clinical team members. The time required to implement evaluation components was estimated by PIs who completed the needs assessment survey and, in some cases, was estimated for service components that they may not have routinely provided. Future implementation studies are needed to assess these components precisely. In addition, not all DCM Consortium site PIs, GCs, or APPs were able to attend the Summer Scientific Symposium. Given time constraints, not all focus groups were able to discuss all questions. Focus group facilitators were GCs, a key genetics professional group in this study; however, the Consolidated Criteria for Reporting Qualitative Research checklist ([Supplemental Material](#)) has provided additional transparency on the positionality of the facilitators. Care model groups had small sample sizes, limiting the strength of model-specific inferences. Despite limitations, the PI responses reported herein provide a broad and clinically relevant set of perspectives, as DCM Consortium PIs had the expertise to critically reflect on the patterns and needs of the practice of genetics care in HF as shared through qualitative analysis. Beyond the scope of the current study was a more in-depth evaluation of disparities in genetics services among patients with DCM and their families, including issues related to race, ethnicity, income, education, access, and other factors that are essential to address for all patients with DCM. This is especially the case for Black patients who have an increased risk of DCM, as previously shown by studies from the DCM Consortium.^{14,15}

Conclusions

Current care models for the provision of genetic services across DCM Consortium sites are highly variable. To address the need for genetic evaluation to be equitably accessible to all patients with DCM, identifying practice standards to implement components of a DCM genetic evaluation that can be adapted and applied across highly variable institutional environments may yield more scalable clinical genetics care models.

ARTICLE INFORMATION

Received April 05, 2026; accepted August 11, 2026.

Affiliations

Ohio State University (E.J., H.K., P.K.P., A.N., G. H., R.E.H.). Stanford University (T.M., N.R., M. K.). Duke University Hospital (P.G.). New York University Langone Health (S.W., S.K.). Cleveland Clinic (J.L., K.G.A., W.H.W.T.). Medical University of South Carolina (N.B., D.P.J.). Johns Hopkins University (E.E.B.). Inova Heart and Vascular Institute (A.B., P.S.). University of Pennsylvania (J.C., A.O.). University of Alberta (S.C.). Emory University (A.E., M.A.B., J.D.). University of California, Los Angeles (J.F., J. W.). Genome Medical (M. Kisling). Cedars-Sinai Medical Center (D.M., E.P.K.). Cincinnati Children's Hospital Medical Center (E.M.M.). Medstar/Washington Hospital Center (J.S., M.H.). University of Texas Southwestern Medical Center (B.R., M.H.D.). University of Minnesota (L.V.W., C.S.). Dilated Cardiomyopathy Consortium Patient Advisory Board (W.F., G.L.H.). Washington University (G.A.E.). University of Maryland (S.S.G.). Tufts University (G.S.H.). South Miami

Hospital (J.J.). Methodist Hospital (C.M.M.). University of Washington (E. Minami). University of Michigan (S.S.). Louisiana State University (F.S.). University of Nebraska Medical Center (D.S.). University of Alabama, Birmingham (J.T.). Northwestern University (J.W.).

Sources of Funding

This work was supported by National Institutes of Health awards R01HL149423 and R01HL148581. The content is solely the responsibility of the authors and does not represent the official views of the National Institutes of Health.

Disclosures

None relevant to this work.

Supplemental Material

Survey Instrument

Focus Group Interview Guide

Tables S1–S4

Figure S1

REFERENCES

- Parikh VN, Day SM, Lakdawala NK, Adler ED, Olivotto I, Seidman CE, Ho CY. Advances in the study and treatment of genetic cardiomyopathies. *Cell*. 2025;188:901–918. doi: 10.1016/j.cell.2025.01.011
- Landstrom AP, Ferguson JF, James CA, Key KV, Lanfear D, Natarajan P, Rasmussen-Torvik LJ, Reza N, Roden DM, Tsao PS, et al. Genetic and genomic testing in cardiovascular disease: a policy statement from the American Heart Association. *Circulation*. 2025;152:e474–e489. doi: 10.1161/CIR.0000000000001385
- Musunuru K, Hershberger RE, Day SM, Klinedinst NJ, Landstrom AP, Parikh VN, Prakash S, Semsarian C, Sturm AC; American Heart Association Council on Genomic and Precision Medicine; Council on Arteriosclerosis, Thrombosis and Vascular Biology; Council on Cardiovascular and Stroke Nursing; Council on Clinical Cardiology. Genetic testing for inherited cardiovascular diseases: a scientific statement from the American Heart Association. *Circ Genom Precis Med*. 2020;13:e000067. doi: 10.1161/HCG.0000000000000067
- Wilde AAM, Semsarian C, Marquez MF, Sepehri Shamlou A, Ackerman MJ, Ashley EA, Sternick EB, Barajas-Martinez H, Behr ER, Bezzina CR, et al; Document Reviewers. European Heart Rhythm Association (EHRA)/Heart Rhythm Society (HRS)/Asia Pacific Heart Rhythm Society (APHRS)/Latin American Heart Rhythm Society (LAHRS) Expert Consensus Statement on the state of genetic testing for cardiac diseases. *Heart Rhythm*. 2022;19:e1–e60. doi: 10.1016/j.hrthm.2022.03.1225
- Hershberger RE, Givertz MM, Ho CY, Judge DP, Kantor PF, McBride KL, Morales A, Taylor MRG, Vatta M, Ware SM. Genetic evaluation of cardiomyopathy—a Heart Failure Society of America practice guideline. *J Card Fail*. 2018;24:281–302. doi: 10.1016/j.cardfail.2018.03.004
- Hershberger RE, Givertz MM, Ho CY, Judge DP, Kantor PF, McBride KL, Morales A, Taylor MRG, Vatta M, Ware SM; ACMG Professional Practice and Guidelines Committee. Genetic evaluation of cardiomyopathy: a clinical practice resource of the American College of Medical Genetics and Genomics (ACMG). *Genet Med*. 2018;20:899–909. doi: 10.1038/s41436-018-0039-z
- Morales A, Moretz C, Ren S, Smith E, Callis TE, Hall T, Hatchell KE, Nussbaum RL, Regalado E, Rojahn S, et al. Real-world genetic testing utilization among patients with cardiomyopathy. *Circ Genom Precis Med*. 2024;17:e004028. doi: 10.1161/CIRCGEN.122.004028
- Hellwig LD, Banaag A, Olsen C, Turner C, Haigney M, Koehlmoos T. A health systems assessment of genetic counseling in cardiovascular care in a large health system: adherence to genetics recommendations in the Military Health System. *J Genet Couns*. 2024;33:888–896. doi: 10.1002/jgc4.1791
- Longoni M, Bhasin K, Ward A, Lee D, Nisson M, Bhatt S, Rodriguez F, Dash R. Real-world utilization of guideline-directed genetic testing in inherited cardiovascular diseases. *Front Cardiovasc Med*. 2023;10:1272433. doi: 10.3389/fcvm.2023.1272433
- Bui QM, Morales A, Adler ED, Urey MA, Sinha A, Wang J, Hong KN, Ramsis M. Real-world genetic testing practices in cardiomyopathy: national insights from 1.7 million patients. *J Card Fail*. 2026;32:1447–1450. doi: 10.1016/j.cardfail.2025.11.496
- Morales A, Goehring J, McDonald PL, Sanoudou D. Implementation of genetic testing for heritable cardiac conditions: a scoping review. *Genet Med Open*. 2025;3:103441. doi: 10.1016/j.gimo.2025.103441

12. Tang WHW, Bui QM, Cirino AL, Dellefave-Castillo L, Floyd BJ, Guerchicoff A, Guerchicoff M, A VK, Knowles JW, Lafayette K, et al. Cardiologists' perceptions of cardiogenetic testing and management. *JACC Adv*. 2025;4:101910. doi: 10.1016/j.jaccadv.2025.101910
13. Kinnamon DD, Morales A, Bowen DJ, Burke W, Hershberger RE; DCM Consortium. Toward genetics-driven early intervention in dilated cardiomyopathy: design and implementation of the DCM precision medicine study. *Circ Cardiovasc Genet*. 2017;10:e001826. doi: 10.1161/CIRCGENETICS.117.001826
14. Huggins GS, Kinnamon DD, Haas GJ, Jordan E, Hofmeyer M, Kransdorf E, Ewald GA, Morris AA, Owens A, Lowes B, et al; DCM Precision Medicine Study of the DCM Consortium. Prevalence and cumulative risk of familial idiopathic dilated cardiomyopathy. *JAMA*. 2022;327:454–463. doi: 10.1001/jama.2021.24674
15. Jordan E, Kinnamon DD, Haas GJ, Hofmeyer M, Kransdorf E, Ewald GA, Morris AA, Owens A, Lowes B, Stoller D, et al; DCM Precision Medicine Study of the DCM Consortium. Genetic architecture of dilated cardiomyopathy in individuals of African and European ancestry. *JAMA*. 2023;330:432–441. doi: 10.1001/jama.2023.11970
16. Jordan E, Ni H, Parker P, Kinnamon DD, Owens A, Lowes B, Shenoy C, Martin CM, Judge DP, Fishbein DP, et al; DCM Consortium. Implementing precision medicine for dilated cardiomyopathy: insights from the DCM consortium. *Circ Genom Precis Med*. 2025;18:e005078. doi: 10.1161/CIRCGEN.125.005078
17. Weiner BJ, Lewis CC, Stanick C, Powell BJ, Dorsey CN, Clary AS, Boynton MH, Halko H. Psychometric assessment of three newly developed implementation outcome measures. *Implement Sci*. 2017;12:108. doi: 10.1186/s13012-017-0635-3
18. Proctor E, Silmere H, Raghavan R, Hovmand P, Aarons G, Bunger A, Griffey R, Hensley M. Outcomes for implementation research: conceptual distinctions, measurement challenges, and research agenda. *Adm Policy Ment Health*. 2011;38:65–76. doi: 10.1007/s10488-010-0319-7
19. Tong A, Sainsbury P, Craig J. Consolidated Criteria for Reporting Qualitative Research (COREQ): a 32-item checklist for interviews and focus groups. *Int J Qual Health Care*. 2007;19:349–357. doi: 10.1093/intqhc/mzm042
20. Damschroder LJ, Aron DC, Keith RE, Kirsh SR, Alexander JA, Lowery JC. Fostering implementation of health services research findings into practice: a consolidated framework for advancing implementation science. *Implement Sci*. 2009;4:50. doi: 10.1186/1748-5908-4-50
21. Glaser BG, Strauss A. *A Discovery of Grounded Theory: Strategies for Qualitative Research*. Aldine; 1967.
22. Mackley MP, Richer J, Guerin A, Caluseriu O, Armstrong L, Blood KA, Bernier F, Boswell-Patterson C, Chard M, Costain G, et al. Mainstreaming of clinical genetic testing: a conceptual framework. *Genet Med*. 2025;27:101465. doi: 10.1016/j.gim.2025.101465
23. Mohananey A, Tseng AS, Julakanti RR, Gonzalez-Bonilla HM, Kruisselbrink T, Prochnow C, Rodman S, Lin G, Redfield MM, Rosenbaum AN, et al. An intervention strategy to improve genetic testing for dilated cardiomyopathy in a heart failure clinic. *Genet Med*. 2023;25:100341. doi: 10.1016/j.gim.2022.11.009
24. Riordan S, Richardson J, Zierhut H, Goodnight BL, Sieling FH, Black CM, Moore RA. Medicare beneficiary barriers to genetic counselor services: implications for patient policy, decision-making, and care. *J Genet Couns*. 2024;33:262–268. doi: 10.1002/jgc4.1729
25. Sturm AC, Knowles JW, Gidding SS, Ahmad ZS, Ahmed CD, Ballantyne CM, Baum SJ, Bourbon M, Carrie A, Cuchel M, et al; Convened by the Familial Hypercholesterolemia Foundation. Clinical genetic testing for familial hypercholesterolemia: JACC scientific expert panel. *J Am Coll Cardiol*. 2018;72:662–680. doi: 10.1016/j.jacc.2018.05.044
26. Isselbacher EM, Preventza O, Hamilton Black J III, Augoustides JG, Beck AW, Bolen MA, Braverman AC, Bray BE, Brown-Zimmerman MM, Chen EP, et al; Peer Review Committee Members. 2022 ACC/AHA guideline for the diagnosis and management of aortic disease: a report of the American Heart Association/American College of Cardiology Joint Committee on Clinical Practice Guidelines. *Circulation*. 2022;146:e334–e482. doi: 10.1161/CIR.0000000000001106
27. Wilde AAM, Semsarian C, Marquez MF, Sepehri Shamloo A, Ackerman MJ, Ashley EA, Sternick Eduardo B, Barajas-Martinez H, Behr ER, Bezzina CR, et al. European Heart Rhythm Association (EHRA)/Heart Rhythm Society (HRS)/Asia Pacific Heart Rhythm Society (APHRS)/Latin American Heart Rhythm Society (LAHRS) Expert Consensus Statement on the state of genetic testing for cardiac diseases. *J Arrhythm*. 2022;38:491–553. doi: 10.1002/joa3.12717
28. Olson M, Anderson J, Knapke S, Kushner A, Martin L, Statile C, Shikany A, Miller EM. Cardiac genetic counseling services: exploring downstream revenue in a pediatric medical center. *J Genet Couns*. 2025;34:e1984. doi: 10.1002/jgc4.1984
29. American College of Obstetricians and Gynecologists' Committee on Practice Bulletins—Obstetrics; Committee on Genetics; Society for Maternal-Fetal Medicine. Screening for fetal chromosomal abnormalities: ACOG Practice Bulletin, Number 226. *Obstet Gynecol*. 2020;136:e48–e69. doi: 10.1097/AOG.0000000000004084
30. American College of Obstetricians and Gynecologists Committee on Genetics Opinion No. 486: update on carrier screening for cystic fibrosis. *Obstet Gynecol*. 2011;117:1028–1031. doi: 10.1097/AOG.0b013e31821922c2
31. American College of Obstetricians and Gynecologists Committee on Genetics Opinion No. 469: carrier screening for fragile X syndrome. *Obstet Gynecol*. 2010;116:1008–1010. doi: 10.1097/AOG.0b013e3181fae884
32. American College of Obstetricians and Gynecologists Committee on Genetics Opinion No. 442: preconception and prenatal carrier screening for genetic diseases in individuals of Eastern European Jewish descent. *Obstet Gynecol*. 2009;114:950. doi: 10.1097/AOG.0b013e3181bd12f4
33. Network NCC. Genetic/familial high-risk assessment: breast, ovarian, pancreatic, and prostate. Accessed November 14 2025. https://www.nccn.org/guidelines/category_2
34. Tayal U, Prasad S, Cook SA. Genetics and genomics of dilated cardiomyopathy and systolic heart failure. *Genome Med*. 2017;9:20. doi: 10.1186/s13073-017-0410-8